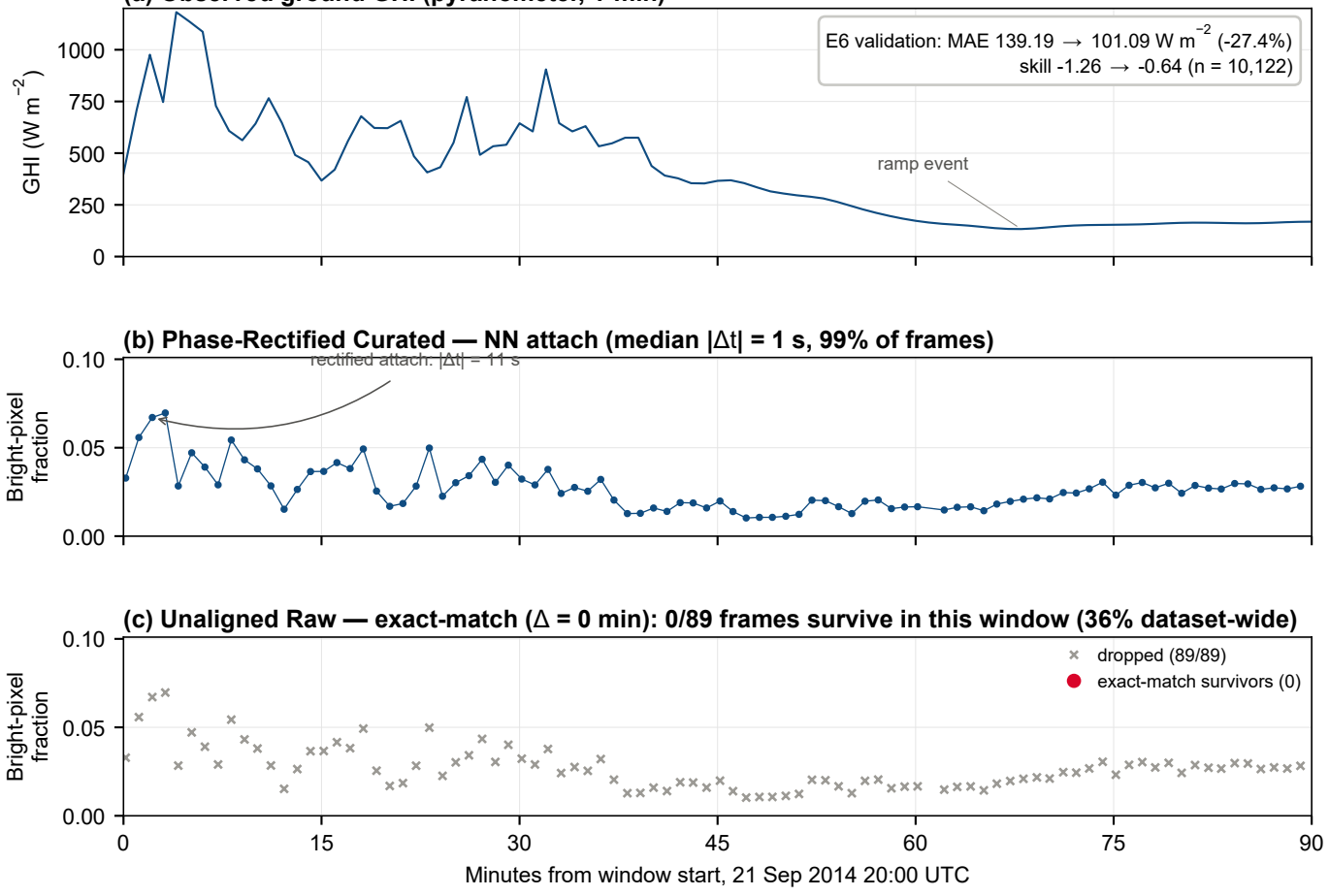

(a) Observed ground GHI (pyranometer, 1-min)
GHI (W m^{-2})
1000
750
500
250
0
E6 validation: MAE 139.19 → 101.09 W m^{-2} (-27.4%)
skill -1.26 → -0.64 (n = 10,122)
ramp event
(b) Phase-Rectified Curated — NN attach (median |Δt| = 1 s, 99% of frames)
Bright-pixel fraction
0.10
0.05
0.00
rectified attach: |Δt| = 11 s
(c) Unaligned Raw — exact-match (Δ = 0 min): 0/89 frames survive in this window (36% dataset-wide)
Bright-pixel fraction
0.10
0.05
0.00
dropped (89/89)
exact-match survivors (0)
0
15
30
45
60
75
90
Minutes from window start, 21 Sep 2014 20:00 UTC